

Tweedie Optimization

M5 Retail Demand Forecasting — Problem Statement 11

Phase 4 — power search and a failed transfer

NPN AIA Hackathon — St. Joseph's College of Engineering

Phase 4 — an eight-point search over the Tweedie variance power. Generated 2026-08-14.

Terms. **RMSE** — average error with big misses punished far more heavily; lower is better. **MAE** — the plain average error. **Leakage** — letting the model see information that would not have existed when the forecast was really made. **Tweedie** — a loss function for data that is never negative and mostly zero. **Objective / loss function** — the definition of "wrong" that the model tries to minimise. **Inner window** — an earlier 28-day period we tune on, so the real scoring window stays untouched.

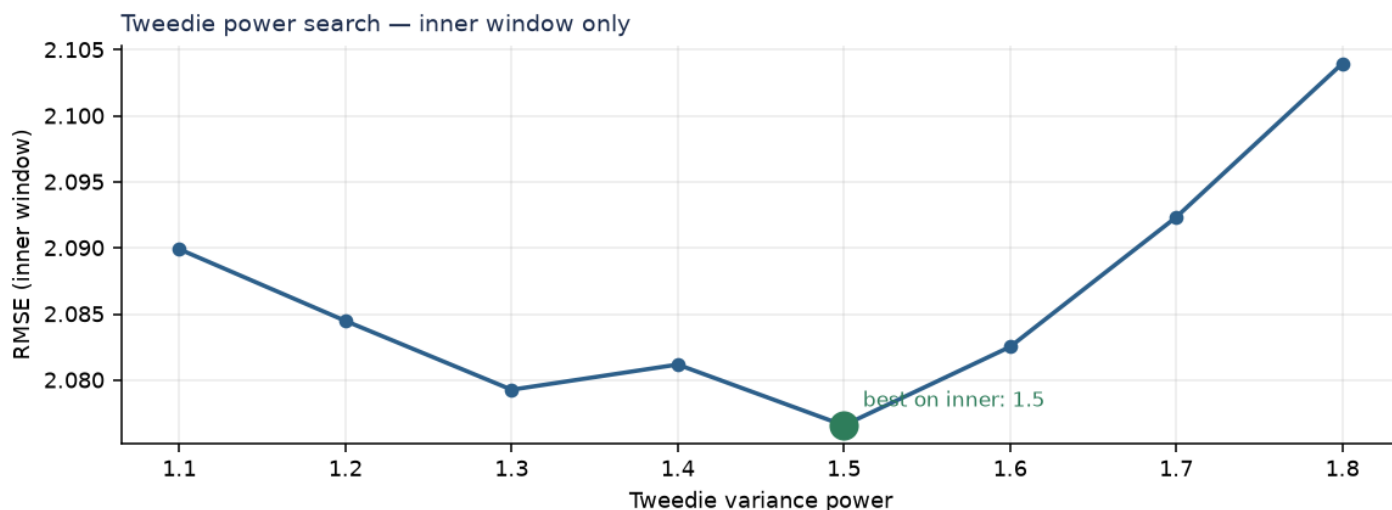
Every number here comes from an experiment that actually ran. Failed experiments are reported alongside successful ones — in this campaign most of them failed, and that is the finding.

What the Tweedie power controls

Tweedie sits between two familiar distributions. A power near 1 behaves like a Poisson (counting things); a power near 2 behaves like a Gamma (positive continuous amounts). In between it can put mass exactly at zero *and* have a long right tail — which is what daily unit sales look like. We had been using 1.1 without ever testing it.

Search on the inner window

Power	Inner RMSE	Inner MAE	High-volume RMSE	High-volume bias
1.1 *(previous setting)*	2.0899	1.0173	5.938	+0.317
1.2	2.0845	1.0147	5.906	+0.297
1.3	2.0793	1.0124	5.896	+0.249
1.4	2.0812	1.0120	5.896	+0.258
1.5	2.0766	1.0087	5.872	+0.129
1.6	2.0825	1.0090	5.902	+0.125
1.7	2.0923	1.0092	5.939	+0.043
1.8	2.1040	1.0096	5.981	-0.063



A clean U-shape with a minimum at **power 1.5**. The high-volume bias falls steadily as the power rises — exactly the behaviour Phase 3 said we wanted.

Then the honest part

The power was selected using only the inner window, so applying it once to the primary window is a clean test. It did not survive:

	Inner window	Primary window
Power 1.1 (previous)	2.0899	2.1210
Power 1.5 (selected)	2.0766	2.1263
Change	-0.0133	+0.0053

A gain of about 0.013 on one window became a loss of about 0.005 on the next. MAE did improve slightly (-0.0030).

Decision: keep power 1.1. The improvement was not real. Had we selected on the scoring window instead of an inner one, we would have shipped noise and called it a result.